

Question	Answer	Marks
2(a)	$(p =) mv$ or 4.0×45 or 2.0×85 or $89v$	C1
	$(4.0 \times 45) - (2.0 \times 85) = 89v$ $v = 0.11 \text{ ms}^{-1}$	A1
2(b)(i)	1. speed of approach = 47 ms^{-1} and 2. speed of separation = 0	A1
2(b)(ii)	speed of separation less than/not equal to speed of approach and so inelastic collision	A1
2(c)	force is equal to rate of change of momentum	B1
	force on ball (by block) <u>equal and opposite</u> to force on block (by ball) so rates of change of momentum are <u>equal and opposite</u>	B1
	or	
	force on ball (by block) <u>equal and opposite</u> to force on block (by ball)	(B1)
	force is equal to rate of change of momentum so rates of change of momentum are <u>equal and opposite</u>	(B1)

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Question	Answer	Marks
2(c)(i)	work (done) / time (taken)	D1